```
function main



V = 0:1:1;
C0 = [40 0 0];
e=1;


while ( e >= 0.01);

[V,C] = ode45(@fode, V, C0);


e= ((C(1)-C(2))^2 + (C(2)-C(3))^2)/2;
C0 = [(v*Caf+v*C(1))/(v+vr); (v*C(2)/(v+vr)) ; v*C(3)/(v+vr)];

end


end



function fode


k1 = 0.001;
k2 = 0.0001;
Caf = 40;
L = 10;
A = 0.01;
v = 1;
vr = 1;

Ca = C(1);
Cb = C(2);
Cc = C(3);

r1 = k1 * Ca^2;
r2 = k2 * Cb;
 R=[r1 , r2];

alpha = [ -2 1 0
    0 -1 1];
% stoichiometric consts from rxn

dc = alpha' * r;


cv(1) = (-k1 * Ca^2)/(v+vr);
cv(2) = (k1 * Ca^2 - k2 * Cb)/(v+vr);
cv(3) = (k2 * Cb)/(v+vr);
```

```
end

function Conversion



VR = [0, 0.1, 0.5, 1, 10, 20, 100];
x=1;
conv = zeros(1,7);

while (x <= 7);



call fode;
vr = VR(x);
conv(x) = C(1)/Caf;


x=x+1;
end

Figure(1)
plot(VR, conv);
xlabel (' recycle flowrate ') ; ylabel (' Conversion of A ');
end
```